TRANSPORTATION OPTIMISATION FOR A LAGOS PHARMACEUTICAL DISTRIBUTOR

Key findings

- The lowest-cost weekly plan costs ₦92,800, confirmed independently in Python and Excel.
- The network has only 40 cartons of spare stock. If Apapa's stock falls below 210 cartons, demand cannot be met at all.
- A disruption on the Apapa→Lekki route is the costliest scenario tested (+₦2,900 a week, +3.1%), while closing Apapa→Surulere costs only +₦800 (+0.9%).

Problem

A hypothetical pharmaceutical distributor in Lagos supplies five pharmacies: Yaba, Surulere, Lekki, Ikorodu and Ajah, from two warehouses, Ikeja and Apapa. Each week it must decide how many cartons to distribute from each warehouse to each pharmacy. Delivery costs differ by route, warehouse supply is limited, and every pharmacy must receive its full weekly demand. This project builds an optimisation model that finds the lowest-cost weekly distribution plan, then tests how that plan holds up under disruptions such as an expensive route, a closed route and a stock shortage.

Data

All figures are illustrative and were created for this exercise. They do not come from a real company. Costs are in naira (₦) per carton.

Delivery cost per carton (₦)	Yaba	Surulere	Lekki	Ikorodu	Ajah	**Supply (cartons)**
Ikeja	120	150	260	200	300	**300**
Apapa	140	110	210	280	230	**250**
Weekly demand (cartons)	**90**	**110**	**130**	**80**	**100**	

Total supply is 550 cartons against total demand of 510, so the network has 40 cartons of spare stock.

Assumptions

1. All products are treated as one item, measured in cartons.
2. The delivery cost per carton is constant on each route, so total cost grows in direct proportion to the quantity distributed (no volume discounts).

3. Weekly demand is known and fixed, and every pharmacy must receive its full demand.
4. Warehouse stock is the only supply limit, and it is available at the start of the week.
5. There are no vehicle capacity limits, route capacity limits or delivery time windows.
6. Costs and supply stay fixed within each scenario.

Method

The problem is a classic transportation problem, formulated as a linear program (LP).

Sets and data: warehouses $i \in$ {Ikeja, Apapa} and pharmacies $j \in$ {Yaba, Surulere, Lekki, Ikorodu, Ajah}. For each route, $c(i,j)$ is the cost per carton. Each warehouse has supply $s(i)$ and each pharmacy has weekly demand $d(j)$.

Decision variables: $x(i,j) \geq 0$, the number of cartons distributed from warehouse i to pharmacy j.

Objective: minimise total delivery cost, $\Sigma\, c(i,j) \cdot x(i,j)$ over all routes.

Constraints:

Supply: for each warehouse, total distributed cannot exceed its supply, $\Sigma_j\, x(i,j) \leq s(i)$.

Demand: for each pharmacy, total received must meet its demand, $\Sigma_i\, x(i,j) \geq d(j)$.

Because total supply (550) exceeds total demand (510), the problem is unbalanced, so the supply constraints are inequalities rather than equalities and some supply may stay unused. The model was solved in Python using the PuLP library (version 3.3.2) and its CBC solver, and cross-checked independently in Excel Solver using the Simplex LP method. Because all data are whole numbers, the transportation structure guarantees an optimal plan in whole cartons, so no integer restriction was needed.

Base results

The lowest-cost weekly plan costs ₦92,800.

Shipments (cartons)	Yaba	Surulere	Lekki	Ikorodu	Ajah	Total out	Supply
Ikeja	90	90	0	80	0	260	300
Apapa	0	20	130	0	100	250	250

Every pharmacy receives exactly its demand. Apapa's stock is fully used, while Ikeja keeps 40 cartons in reserve. Surulere is the only pharmacy supplied from both warehouses.

Validation

The model was checked in four ways.

1. It was solved independently in Excel Solver, giving the same cost (₦92,800) and the same shipments as Python.
2. The plan was verified by hand: each warehouse distributes no more than its stock, each pharmacy receives its demand, and the cost recalculates to ₦92,800.
3. The solver status was checked on every run; the status must read "Optimal".

Scenario analysis

Scenario	Change	Status	Cost (₦)	vs base	Routes that changed
A	Apapa→Lekki cost rises from 210 to 320	Optimal	95,700	+2,900	Lekki moves to Ikeja (0→130 from Ikeja); this fills Ikeja, so Surulere moves entirely to Apapa (Apapa→Surulere 20→110, Ikeja→Surulere 90→0)
B	Apapa→Surulere route closed	Optimal	93,600	+800	Ikeja covers all of Surulere (90→110)
C	Apapa supply falls from 250 to 220	Optimal	94,100	+1,300	Ikeja supplies 10 cartons to Lekki; Surulere comes entirely from Ikeja
D	Apapa supply falls to 150	Infeasible	n/a	n/a	Total stock (450) is below demand (510)

The network holds only 40 cartons of spare stock. The plan stays feasible as long as Apapa has at least 210 cartons (at exactly 210, cost is ₦94,600 and there is no spare stock left).

Recommendations

1. Set a minimum supply level of 210 cartons at Apapa, with a reorder alert at about 230. Below 210 the network cannot meet demand at all, and the cost climbs from ₦92,800 to ₦94,100 at 220 cartons and ₦94,600 at 210 cartons.
2. Prepare a backup plan for the Lekki route before it's needed. A disruption on Apapa→Lekki (scenario A) costs an extra ₦2,900 a week (+3.1%) and forces a reshuffle: Lekki moves to Ikeja, which then fills to its 300-carton limit, so Surulere has to be served entirely from Apapa. Dispatchers should know this in advance.

3. Don’t spend money protecting the Apapa→Surulere route. Closing it costs only ₦800 a week (+0.9%), because Ikeja can absorb the demand. Resilience spending is better aimed at Apapa’s supply level and the Lekki route.

Limitations

The data used are illustrative, so the numbers show how the method behaves rather than predicting a real distributor’s costs. The model assumes costs are linear in the quantity supplied. Trucks have fixed trip costs and limited capacity, which can make a plan that is optimal here impractical. Pharmaceutical products also add real constraints that this model ignores, such as expiry dates, cold-chain requirements and multiple product types. Demand is treated as known and fixed, although in practice it varies from week to week.

REFERENCE

https://colab.research.google.com/drive/1VMVYfq6VwYrXtTwmxtDRu-7pzIboeG0e?usp=sharing